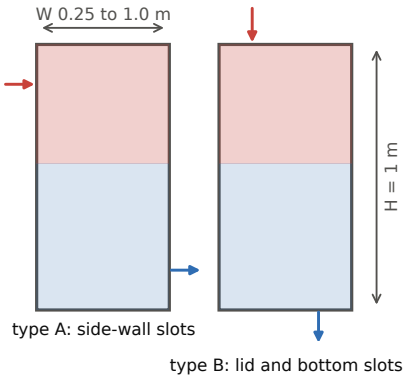
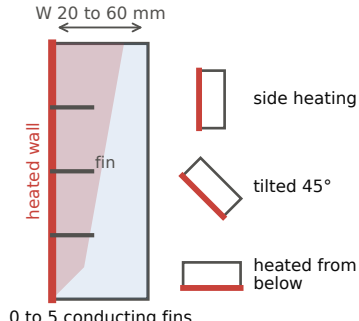
(a) Stratified water tank (sensible)

1,212 runs; real water properties;
laminar and k-ω SST



(b) PCM melting cell (latent)

500 runs; n-octadecane
and lauric acid



(c) Test splits: what is held out of training

Tank	
random	10 % of design points
unseen width	W outside 0.33 to 0.93 m
discharge	all discharging runs (trained on charging)
faster inflow	Re above the design 90th percentile (SST)
PCM	
random	10 % of cases
held-out fins	4 and 5 fins (trained on 0 to 3)
stronger heating	superheat above the 90th percentile
unseen tilt	45° and heated from below
material transfer	lauric acid, trained on n-octadecane
(reverse)	n-octadecane, trained on lauric acid

Splits are by design point; laminar and SST twins never straddle train and test.